

Discovering efficient gate sets for fault-tolerant quantum computation

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We introduce SWIFTBOT, a pipeline for discovering efficient gate sets for fault-tolerant quantum computation. It operates in two modes. In *universal* mode, it ranks $(\mathcal{C}, g)$ pairs — a finite base group $\mathcal{C} \subset \text{SU}(d)$ extended by a single gate g — by the Quantum Circuit Overhead Q_T , the asymptotic cost of approximating Haar-uniform targets. In *application-specific* mode, it minimizes the mean synthesis depth over a structured target family picked out by the application. A third stage matches surviving pairs against a curated catalog of transversal QEC codes, magic-state distillation/cultivation/code-switching protocols, and certified exact-synthesis rings, flagging which pairs admit an end-to-end fault-tolerant stack. At $d = 3$ the universal mode yields the first systematic Q_T ranking across three Σ -series subgroups, with arbitrary-SU(3) extensions on the largest approaching the Kesten–McKay floor within 32%, while structured diagonal and Howard–Vala extensions sit 3–70 $\times$ above. In application-specific mode, a non-canonical rational-phase extension beats canonical Clifford+ T by 6.8 $\times$ on a lattice-gauge target family, showing that the best general-purpose and best target-specific extensions can differ by nearly an order of magnitude. The pipeline, registered groups, fault-tolerance catalog, and reproduction data are released with the paper.

I. INTRODUCTION

Fault-tolerant quantum computation requires a finite universal gate set whose members have cheap physical realizations. The tension between these requirements is sharp: the Eastin–Knill theorem [1] forbids any non-trivial quantum error-correcting code from admitting a universal *transversal* gate set. Practical designs therefore split the gate set into a transversal subgroup $\mathcal{C} \subset \text{SU}(d)$ that the error-correcting code protects cheaply, plus one or more non-transversal extension gates g made fault-tolerant by any of several routes: magic-state distillation (MSD) [2, 3], magic-state cultivation [4], code switching [5], or beyond-stabilizer codes with native non-Clifford transversal action [6–8]. Each extension gate g comes with its own cost: a dedicated magic state, a code that supports it transversally, a distillation or cultivation circuit, or some combination of the above.

Why qudits. The qubit version of this story has a standard resolution: Clifford+ T [2, 9], with Clifford transversal on Steane, Reed–Muller, and Bravyi–Haah families and T distilled via triorthogonal MSD [2, 3]; single-qubit synthesis into this gate set is solved with certified T -count bounds [10, 11]. This toolkit is mature but not closed: as we show for the binary icosahedral base $2I$ (Section IV), systematic $(\mathcal{C}, g)$ exploration beyond the canonical choice remains productive even at $d = 2$, and four developments motivate extending the search into qudit registers at $d > 2$.

Qudit hardware has matured on several fronts. Trapped-ion platforms have demonstrated single-qudit registers up to $d = 7$ [12] and operate multi-ion chains at lower d ; superconducting cavities encode $\gtrsim 10$ Fock levels

per mode with coherence times approaching the best qubit counterparts [13]; reconfigurable neutral-atom arrays provide individually addressable sites at the hundred-qudit scale [14]; and solid-state spin- S systems offer $d = 2S + 1$ native encodings [15, 16]. The binary-encoding overhead qubits impose on inherently higher- d physics is a cost, not a feature of these architectures.

Such hardware development has been matched by advances in theoretical proposals for fault tolerance, both on the QEC side and on the non-transversal-gate side.

Qudit codes have progressed beyond the qubit toolkit. Gottesman’s prime- d stabilizer construction [17] spelled out the foundational split from the qubit case; beyond stabilizer, the *nonadditive* Kubischta $((7, 2, 3))_{2I}$ code [6] and Denys–Leverrier *covariant* $2T/2O$ codes [7, 8] carry transversal non-Clifford groups that sidestep the Anderson–Jochym–O’Connor (AJOC) qubit-stabilizer no-go [18]. HEP-motivated bosonic proposals such as the $2T$ -quocit encode specific discrete subgroups natively at $d = 8$ [19]. Non-local qLDPC memories like the bivariate-bicycle code [20] need connectivity graphs whose natural hardware substrates — the same platforms above — make qudit qLDPC codes [21, 22] a convergence rather than two separate programs.

Distillation and cultivation are advancing in parallel. Qutrit triorthogonal distillation [23, 24] reaches $\gamma \approx 1.51$, beating qubit triorthogonal codes below ~ 300 physical qudits; asymptotically-good qudit constructions [25, 26] deliver constant-overhead MSD. Magic-state cultivation [4, 27, 28] is currently qubit-native and surface-code-specific, and an open qudit analogue would close one of the two remaining arguments for $d = 2$ canonicalization (Section VIII.1).

Applications drive the practical case for qudits. Lattice-gauge simulations of $\text{SU}(N)$ live on $|\mathcal{C}|$ -dimensional per-link registers that fit qudit architectures natively $(|\Sigma(36 \times 3)| = 108 \text{ states per link, encoded on three qutrits})$

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plus two qubits [29, 30]); experimental trapped-ion realizations of $U(1)$ and $SU(2)$ gauge dynamics already use native qudits [31, 32], and native-qudit QEC constructions are in active development for $\mathbb{Z}_N$ [33] and $SU(2)$ [34] lattice gauge theories. Beyond lattice gauge, higher- d registers reduce the Trotter cost of continuous-variable bosonic simulations, cut the ancilla overhead of fermion-orbital encodings in chemistry and materials science, and absorb large-spin degrees of freedom (magnetism, $S > \frac{1}{2}$ impurities) without binary unpacking [15, 16].

qLDPC codes and qudit extensions. A complementary insight: modern codes like qLDPC may be better understood as qudit codes than as collections of qubit codes. The gate set accessible on a $[[n, k, d]]$ qLDPC code like the $[[144, 12, 12]]$ gross code [20] through fold-transversal Cliffords and code automorphisms is *not* the product group $\text{Clifford}(2)^{\otimes k}$: recent work on bivariate-bicycle codes shows the fold-transversal Cliffords act collectively on the logical qubits [35], with the general code-automorphism-to-logical-gate construction [36] and the remaining-logical-Cliffords syndrome-extraction cost [37] now spelled out. Built from code-structured Pauli-string operators, the accessible set is more naturally read as a specific finite subgroup $\mathcal{C} \subset SU(2^k)$ — a single-qudit object at $d = 2^k$ rather than a tensor product of qubit gates. The corresponding non-Clifford completion could then be a single unitary g on the k -qubit logical space: a high-weight entangling gate in the qubit basis, but a single-qudit extension at $d = 2^k$. Distilling the one $|\psi_g\rangle$ magic state is structurally distinct from distilling k separate $|T\rangle$'s, and potentially cheaper: constant-rate qudit distillation [25, 26] makes this a live possibility rather than a formal rephrasing. The $(\mathcal{C}, g)$ ranking this paper formalizes applies once $\mathcal{C}$ is concretely identified and the relevant t -design irreps at $d = 2^k$ are tractable; a gross-code case study is left to future work.

No comparable consensus exists at $d > 2$. A three-parameter family $HV(z, \gamma, \varepsilon)$ of qudit T -gates at prime d is known [38], but no systematic comparison of members has been done. The finite-subgroup classification of $SU(d)$ is complete for $d \leq 4$ [39–43] but open beyond, and transversal-gate classifications for stabilizer codes are available for qubit stabilizer codes [18]; qudit analogues exist for specific code families but not in the same exhaustive form. Prime- d distillation protocols exist [23], qudit synthesis for Shor-class algorithms has been studied [44], and Ross–Selinger-style exact synthesis has recently been extended to Clifford+ R at $d = 3$ [45]. What is missing is a quantitative ranking: given a base group $\mathcal{C}$ and an extension g , how efficiently does $(\mathcal{C}, g)$ synthesize target unitaries?

Slowik, Dulián, and Sawicki [46] supplied the right figure of merit. The *Quantum Circuit Overhead* $Q_T(\mathcal{C}, g)$ is the asymptotic exponent in the word length of an ε -accurate approximation: the number of gates scales as $(1/\varepsilon)^{Q_T}$, bounded below by the Kesten–McKay optimum Q_{opt} and saturated by the “golden gates” of Sarnak [47].

This paper uses Q_T to rank candidates. We introduce

SWIFTBOT, an agentic workflow that enumerates $(\mathcal{C}, g)$ pairs — in principle for any dimension d , tested in the present paper at $d \leq 4$ where classification bounds are known — screens each for universality via the Sawicki–Karnas criterion, evaluates Q_T numerically [46], and cross-references surviving pairs against a curated catalog of 25 transversal QEC codes (stabilizer, beyond-stabilizer, and intrinsic), 10 magic-state distillation / cultivation / code-switching protocols spanning five target-gate families, and two certified exact-synthesis rings.

We make three contributions. First, at $d = 2$ we reproduce the Slowik *et al.* numerics at the percent level (Section IV). Second, we establish the first systematic Q_T ranking at $d = 3$ across the three irreducible-adjoint Σ -series subgroups with $|\mathcal{C}| \in \{216, 648, 1080\}$, using Howard–Vala, rational-phase, and arbitrary- $SU(3)$ extensions (Section V). Third, applied to a gate set relevant for lattice-gauge theory simulation [19, 29, 48], a non-canonical rational-phase extension beats Clifford+ T by $6.8\times$ in mean synthesis depth (Section VII). Together these show that the best general-purpose extension and the best target-specific extension can differ by nearly an order of magnitude—a distinction invisible when Q_T alone is reported.

Code and data are released with this paper [49]. Section II reviews the QCO framework, Section III sketches the pipeline, Sections IV, V and VII present the results, and Section VIII closes with a fault-tolerance audit and open problems.

II. BACKGROUND

We briefly review the three ingredients used throughout: the Quantum Circuit Overhead (QCO) figure Q_T [46] as our primary figure of merit, the Sawicki–Karnas criterion [50] for ruling out non-universal extensions without running QCO, and the Howard–Vala family [38] and discrete-subgroup-approximation (DSA) primitive-gate series [29, 48] as the principal sources of candidate extensions at $d \geq 3$.

II.1. δ -approximate t -designs and Q_T

The Quantum Circuit Overhead Q_T is a property of a *generating set* $\mathcal{S} = \mathcal{C} \cup \{g, g^{-1}\}$, not of the base group $\mathcal{C}$ alone; a poor choice of extension g can make the Cayley random walk converge slowly on $SU(d)$ even on a base group with a large spectral gap. We therefore never quote Q_T without specifying the extension.

Let $\mathcal{S} = \mathcal{C} \cup \mathcal{C}^{-1} \cup \{g, g^{-1}\} \subset SU(d)$ be a symmetric generating set consisting of a finite subgroup $\mathcal{C}$ and a single extension g . Consider the t -design operator on the t -th tensor power of the defining representation,

$$T_t(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \pi_t(s), \quad \pi_t = \bigoplus_{\lambda \in \Lambda_t} \pi_\lambda^{\oplus m_\lambda}, \quad (1)$$

TABLE I. Kesten–McKay floor $Q_{\text{opt}} = \log |\mathcal{C}| / \log(1/\delta_{\text{opt}})$ for the finite subgroups referenced in this paper, with $\delta_{\text{opt}} = 2\sqrt{|\mathcal{C}| - 1/|\mathcal{C}|}$.

d Group	$ \mathcal{C} $	Q_{opt}
2 hurwitz (projective A_4)	12	4.19
2 $\text{Clifford}_1 / 2T$	24	3.47
2 $2O$ (binary octahedral)	48	3.09
2 BI (binary icosahedral $2I$)	120	2.81
3 $\Sigma(60) \simeq A_5$	60	3.00
3 $\Sigma(36 \times 3)$ (Valentiner)	108	2.83
3 $\Sigma(72 \times 3)$	216	2.69
3 $\Sigma(216 \times 3)$ (= qutrit Clifford_1)	648	2.54
3 $\Sigma(360 \times 3)$	1080	2.49

where Λ_t is the set of $\text{SU}(d)$ highest weights that appear in the decomposition of $(\mathbb{C}^d)^{\otimes t} \otimes (\mathbb{C}^d)^{* \otimes t}$ and π_λ is the associated irreducible representation. The t -design spectral radius is

$$\delta_t(\mathcal{S}) = \max_{\lambda \in \Lambda_t \setminus \{0\}} \|T_t(\mathcal{S})\|_{\pi_\lambda}, \quad (2)$$

the largest spectral norm across non-trivial irreps. A smaller δ_t means the random walk generated by $\mathcal{S}$ equidistributes faster on $\text{SU}(d)$.

For an ε -net of $\text{SU}(d)$ in the operator norm, Słowiak *et al.* [46] show that the number of Solovay–Kitaev-style words required scales as $(1/\varepsilon)^{Q_T}$, with

$$Q_T(\mathcal{S}) = \frac{\log |\mathcal{C}|}{\log(\delta_t^{-1}(\mathcal{S}))}, \quad (3)$$

in the limit $t \rightarrow \infty$. The Kesten–McKay bound [51–53] for a k -regular Cayley graph gives, in the $|\mathcal{C}| \rightarrow \infty$ asymptotic we use throughout,

$$\delta_t \geq \delta_{\text{opt}} = \frac{2\sqrt{|\mathcal{C}| - 1}}{|\mathcal{C}|}, \quad (4)$$

and therefore

$$Q_T \geq Q_{\text{opt}} = \frac{\log |\mathcal{C}|}{\log(1/\delta_{\text{opt}})}, \quad (5)$$

with equality for “golden gates” [47]. Table I lists Q_{opt} for every base group in our registry that we use as a baseline below.

In practice Q_T is estimated at finite t by sampling the ensemble in Eq. (1) with a Monte-Carlo extension gate g (the *rnd* baseline) or a deterministic one, then evaluating Eq. (2) via sparse spectral decomposition on each irrep of π_t . The feasibility ceiling at $d = 3$ is set by representation dimensions [46]: the largest irrep at $t = 50$ has dimension $\sim 10^5$, whereas $t = 15$ peaks at $\sim 4 \times 10^3$ and is tractable. At $d = 2$, the Słowiak *et al.* $Q_T(t)$ convergence curves [46] reach stabilization at the percent level by $t \gtrsim 30$. At $d = 3$ our reference point is weaker: we observe the $t = 5$ and $t = 10$ δ values agree

to four decimals for all structured extensions (Section V), which strongly suggests stabilization, but without an independent $t \rightarrow \infty$ reference we cannot tightly bound the residual Q_T error at $d = 3$. An asymptotic extrapolation $Q_T(t \rightarrow \infty)$ from a $Q_T(t)$ sweep is a natural next step. We quote Q_{opt} alongside Q_T throughout to expose the gap to optimality.

II.2. Pre-screening with the Sawicki–Karnas criterion

Not every candidate extension needs a QCO run. Sawicki and Karnas [50] proved that $\mathcal{S}$ generates a dense subgroup of $\text{SU}(d)$ if and only if

- (i) the adjoint representation of $\mathcal{S}$ is irreducible, and
- (ii) $\langle \mathcal{S} \rangle$ — the subgroup generated by $\mathcal{S}$ — is not contained in any finite subgroup of $\text{SU}(d)$ (checked against the Blichfeldt classification caps).

The Sawicki–Karnas criterion is stated on $\mathcal{S}$, but the irreducibility of $\text{Ad}_{\mathcal{S}}$ has a convenient sufficient check on $\mathcal{C}$ alone. Since $\mathcal{C} \subset \mathcal{S}$, every matrix commuting with Ad_s for all $s \in \mathcal{S}$ also commutes with Ad_c for all $c \in \mathcal{C}$, so $\text{Comm}(\text{Ad}_{\mathcal{S}}) \subseteq \text{Comm}(\text{Ad}_{\mathcal{C}})$; if $\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) = 1$ already, then $\dim \text{Comm}(\text{Ad}_{\mathcal{S}}) = 1$ for every g . Condition (i) is therefore *automatic* whenever $\text{Ad}_{\mathcal{C}}$ is irreducible. Schur’s orthogonality relation reduces the check on $\mathcal{C}$ to an $\mathcal{O}(d^2)$ -memory character sum,

$$\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} |\chi_{\text{Ad}}(c)|^2, \quad (6)$$

where $\chi_{\text{Ad}}(c) = \text{tr } \text{Ad}_c$. When $\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) > 1$, condition (i) on $\mathcal{S}$ can still hold if g mixes $\mathcal{C}$ ’s Ad -invariant subspaces; in this case we re-check on the full generating set $\mathcal{C} \cup \{g, g^{-1}\}$ via the joint kernel of the $(\text{Ad}_s \otimes I - I \otimes \text{Ad}_s^\top)$ stack, valid on non-closed sets. Bases with reducible $\text{Ad}_{\mathcal{C}}$ (e.g. $\Sigma(36 \times 3)$, $\Sigma(60) \simeq A_5$, the $s_{60 \times 4}$ family) are not automatically excluded; we defer them in this paper for compute reasons (Section V) but flag them as natural targets for a follow-up. Condition (ii) is enforced at finite cost by running a bounded closure $\langle \mathcal{S} \rangle \cap \text{Cap}(n)$ against the classification of finite subgroups of $\text{SU}(d)$ (orders ≤ 120 at $d = 2$, ≤ 1080 at $d = 3$, ≤ 5040 at $d = 4$) [39–43]; we use cap sizes $n = 120, 1080, 5040$ respectively. An extension is flagged **finite** if closure terminates below the cap, **universal_likely** if closure survives the cap and condition (i) holds, and **infinite_but_not_dense** if condition (i) fails but closure is open.

II.3. Candidate extensions

At $d = 2$ the canonical extension of the Clifford group is $T = P(\pi/4) = \text{diag}(1, e^{i\pi/4})$, giving the Clifford+ T set with certified synthesis via Ross–Selinger [10] and

Kliuchnikov–Maslov–Mosca [11]; different base groups call for different extensions — the Parzanchevski–Sarnak super-golden gates T_{24} on Clifford and T_{12} on the projective A_4 base [54] saturate Q_{opt} at $d = 2$ on their respective bases.¹ We also consider diagonal $P(\theta)$ for $\theta \in \{\pi/8, 2\pi/9, \pi/9, \pi/18\}$ and *arbitrary-SU(d)* extensions — unconstrained single-qudit unitaries sampled uniformly from the (bi-invariant) Haar measure on $\text{SU}(d)$, which we use as a generic-extension baseline; we label these rows **rnd** in the tables below.

At prime d the Howard–Vala family [38] generalizes T via three integers $(z, \gamma, \varepsilon) \in \mathbb{Z}_d^3$. For example at $d = 3$ the $\det = 1$ condition requires a global $\omega^{1/3}$ twist, so the diagonal entries live on 9th (not 3rd) roots of unity [38, Eq. 27]:

$$\text{HV}(z, \gamma, \varepsilon) = \mathcal{N} \cdot \text{diag}(1, \zeta_9^{v_1}, \zeta_9^{v_2}), \quad (7)$$

with $\zeta_9 = e^{2\pi i/9}$, $v_1 = (6z + 2\gamma + 3\varepsilon) \bmod 9$, $v_2 = (6z + \gamma + 6\varepsilon) \bmod 9$, and $\mathcal{N}$ a phase fixing $\det \text{HV}(z, \gamma, \varepsilon) = 1$. (For primes $p > 3$ the base root is ζ_p and v_k is the cubic-in- k recurrence solution $v_k = (k/12)[\gamma + k(6z + (2k-3)\gamma)] + k\varepsilon \pmod{p}$ [38, Eq. 24].) Our panel uses $(z, \gamma, \varepsilon) \in \{(1, 1, 0), (1, 2, 0), (2, 1, 0)\}$: the first is Campbell’s magic gate [23], the second is the Howard–Vala Eq. (27) variant (Clifford-equivalent to the first at $d = 3$ [38] — hence identical δ in Table IV), and the third is a non-Clifford-equivalent orbit. The full group $\{U_v\}$ at $d = 3$ factorizes as $\mathbb{Z}_9 \times \mathbb{Z}_3$ modulo phases [38, Eq. 35]; modulo Clifford conjugation, the non-trivial extensions fall into two orbits (the $\gamma = 0$ orbit lies inside Clifford and is screened out at Stage 2 as a trivial extension), which a direct δ -scan at $t = 5$ on $\Sigma(216 \times 3)$ confirms: all (z, γ, ε) with $\gamma \neq 0$ give $\delta \in \{0.6764, 0.5000\}$, and our panel samples both.

III. SWIFTBOT PIPELINE

SWIFTBOT is a five-stage agentic pipeline built around a heavily modified version of the Słowik–Dulián–Sawicki QCO kernel [46]: we retain the t -design δ -evaluation core but add on-disk $\pi_\lambda(\mathcal{C})$ caching, a conjugation-based rewriting that replaces $|\mathcal{C}|$ exponentials with one, a Python subprocess driver with a persistent SQLite cache, and the Sawicki–Karnas pre-screen, QEC-code catalog, and distillation-protocol catalog that surround the numerics. Numerical stages are deterministic; an LLM supervisor intervenes only at stage boundaries to rank candidates, propose extensions, and decide when to stop. All headline numbers below come from deterministic panels that bypass the LLM; the agent is used for exploratory search not reported here. Full code and a command-line driver

are released with the paper [49]. Fig. 1 summarizes the stage-to-stage flow.

Stage 1 (group registration) computes projective closures $\langle \mathcal{G} \rangle \subset \text{SU}(d)/Z(\text{SU}(d))$ from seed unitaries and stores each group under a content-addressable key. The registry covers the canonical $d \in \{2, 3, 4\}$ subgroups used here: Pauli–Clifford, the projective $A_4 \subset \text{PU}(2)$ (registered as **hurwitz**), and $2I$ (BI) at $d = 2$; the $\Sigma(m \times 3)$ series for $m \in \{36, 72, 216, 360\}$ at $d = 3$; and selected Fairbairn subgroups at $d = 4$.

Stage 2 (universality screen) evaluates $\dim \text{Comm}(\text{Ad}_{\mathcal{C}})$ via Eq. (6) and runs a bounded closure against the classification caps $\{120, 1080, 5040\}$ [39–43]. Pairs whose closure terminates below cap are tagged **finite** and skipped; reducible-adjoint pairs are tagged diagnostically.

Stage 3 (QCO numerics) invokes the kernel of [46] as a subprocess. Two optimizations help: (i) every gate has the form $gcg^\dagger$ with $c \in \mathcal{C}$, so $\pi_\lambda(gcg^\dagger) = \pi_\lambda(g)\pi_\lambda(c)\pi_\lambda(g)^\dagger$ replaces $|\mathcal{C}|$ exponentials with one; (ii) the $\pi_\lambda(c)$ lists are persisted to disk keyed by a fingerprint of $\mathcal{C}$ and the highest weight λ . Together these give 3–6 $\times$ speedup at $d = 3$. Results are stored as per-sample QTRecords in a SQLite cache with full provenance.

Stages 4–5 (fault-tolerance) match each surviving $(\mathcal{C}, g)$ against a curated catalog of QEC codes (Steane, Reed–Muller, Bravyi–Haah, Kubischta $2I$, qutrit triorthogonal, ternary Golay, Pollatsek–Ruskai PI , Kubischta–Teixeira PI , and more) and distillation protocols (Bravyi–Kitaev, Bravyi–Haah, Campbell–Anwar–Browne, programmable MSD, Ouyang–Jing–Brennen (OJB) code-switching). The catalog is curated for tractability, not exhaustive; Hastings–Haah sublogarithmic-overhead codes [55] and the full Litinski resource landscape [56] are notable out-of-scope omissions that would be natural additions. Unmatched pairs return a *research-needed* marker with an EC Zoo [57] pointer rather than guessing.

IV. VALIDATION AT $d = 2$

We validate the SWIFTBOT QCO kernel against the $d = 2$ reference dataset of Słowik, Dulián, and Sawicki [46] on two qubit bases: the Clifford group ($|\mathcal{C}| = 24$) and the 12-element projective $A_4 \subset \text{PU}(2)$ (registered **hurwitz**), each paired with $P(\pi/4)$ and with arbitrary-SU(2) extensions. We also run a systematic 10^4 -sample search on the binary icosahedral group $2I$ (BI, $|\mathcal{C}| = 120$), which carries the Kubischta nonadditive transversal code [6]. All runs are at $t = 50$, matching the reference. A zeroth-order check—replaying the δ -parser on all 5050 rows of the reference file and recomputing Q_T —matches the reference to $\mathcal{O}(10^{-15})$, isolating any later disagreement to the spectral kernel. Headline results are summarized in Table II: Clifford+ $P(\pi/4)$ yields $Q_T = 51.9$ (reference ≈ 52); arbitrary-SU(2) extensions agree at the few-percent level.

Two remarks. First, Clifford+ T has Q_T roughly 14 $\times$ larger than the arbitrary-SU(2) baseline (51.9 vs 3.79)—i.e. its approximation-exponent overhead is worse by an

¹ Explicit matrix forms (SU(2)-normalized): $T_{24} \propto \begin{pmatrix} -1-\sqrt{2} & 2-\sqrt{2}+i \\ 2-\sqrt{2}-i & 1+\sqrt{2} \end{pmatrix}$, $T_{12} \propto \begin{pmatrix} 3 & 1-i \\ 1+i & -3 \end{pmatrix}$; both live in $\mathbb{Z}[i, \sqrt{2}]$.

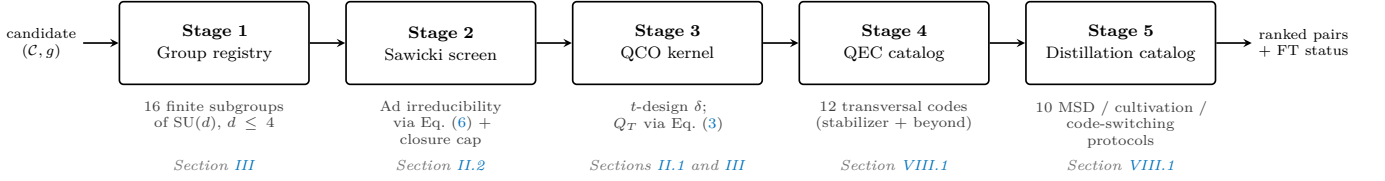


FIG. 1. The five-stage SWIFTBOT pipeline. A candidate $(\mathcal{C}, g)$ pair flows left to right: registry $\rightarrow$ universality screen $\rightarrow$ QCO numerics $\rightarrow$ QEC-code catalog $\rightarrow$ distillation-protocol catalog. Stages 1–3 score the universal-mode Q_T ; Stages 4–5 cross-reference the result against curated fault-tolerance infrastructure. Application-specific mode (Section VII) replaces Stage 3’s Haar-uniform Q_T with a target-family mean synthesis depth on the same $(\mathcal{C}, g)$ pair.

TABLE II. $d = 2$ validation of the SWIFTBOT QCO kernel against the reference dataset of Słowik, Dulián, and Sawicki [46], all at $t = 50$. The Clifford+ $P(\pi/4)$ row is deterministic; rnd rows report the best Q_T across 20 Haar- $SU(2)$ samples for Clifford and hurwitz, and 10^4 samples for BI. Q_{opt} is the Kesten–McKay floor of Eq. (5); BI has no published QCO reference.

Base ($ \mathcal{C} $)	Ext.	Q_T		Q_{opt}
		this work	[46]	
Clifford (24)	$P(\pi/4)$	51.91	≈ 52	3.47
Clifford (24)	rnd	3.79	≈ 3.7	3.47
hurwitz (12)	rnd	4.34	≈ 4.0	4.19
BI (120)	rnd	3.57	—	2.81

order of magnitude; commensurability between $P(\pi/4)$ and the Clifford orbit slows the random walk [46]. This is a Q_T claim, not a verdict on Clifford+ T itself: when the target is structured (Section VII), the ranking inverts. Second, the BI ratio $Q_T/Q_{\text{opt}} \approx 1.27$ at 10^4 samples is far closer to saturation than Clifford’s and arrives with a cataloged fault-tolerant code [6]; the empirical δ distribution on BI has median 0.38 ($Q_T \simeq 4.97$) and 5th-percentile 0.31, and a 2000-resample bootstrap on the best-of- 10^4 Q_T gives a tight 95% CI of [3.57, 3.69]. The best-of- 10^4 floor is the natural super-golden candidate predicted by [47].

BI with structured rational diagonals. The BI super-golden number above is a *Haar* ratio; it does not tell us whether any single deterministic $P(\theta)$ extension is competitive. We probed this with a six-angle rational diagonal panel on BI at $t = 10$ (Table III). Five of the six — $\theta \in \{\pi/4, \pi/8, 2\pi/9, \pi/9, \pi/18\}$ — show the same commensurability slowdown as Clifford+ T , with Q_T/Q_{opt} ratios from 7.8 to 167. The exception is $P(2\pi/5)$, which gives $Q_T = 7.75$, ratio 2.76 to the floor. The pattern is structurally sensible: $2I$ has a 5-fold symmetry inherited from the icosahedron, so $2\pi/5$ is the rational angle *transverse* to the orbit, while all the cyclotomic-8th and 9th/18th-root angles align with subgroup commensurabilities. BI+ $P(2\pi/5)$ is therefore a candidate *deterministic* super-golden gate at $d = 2$: within a factor 2.2 of the Haar best-of- 10^4 , exactly synthesizable in $\mathbb{Z}[\omega_{10}]$, and inherits the Kubischta $((7, 2, 3))$ transversal code on the base.

TABLE III. Binary icosahedral BI ($|\mathcal{C}| = 120$) with rational-angle diagonal extensions $P(\theta)$ at $t = 10$, single-shot. $Q_{\text{opt}} = 2.81$. Only $P(2\pi/5)$ avoids the icosahedral commensurability slowdown.

θ	δ	Q_T	Q_T/Q_{opt}
$\pi/4$ (canonical T)	0.805	22.04	7.84
$\pi/8$	0.949	91.93	32.7
$2\pi/9$ (DSA target)	0.844	28.23	10.0
$\pi/9$	0.960	116.7	41.5
$\pi/18$	0.990	470.3	167
$2\pi/5$	0.539	7.75	2.76

V. $d = 3$ SURVEY

With the SWIFTBOT kernel validated at $d = 2$ (Section IV), we turn to the first systematic Q_T ranking at $d = 3$. The survey covers the three irreducible-adjoint Σ -series subgroups of $SU(3)$ — $\Sigma(72 \times 3)$, $\Sigma(216 \times 3)$, and $\Sigma(360 \times 3)$ —each extended by a common panel of seven candidate gates. (We use the $\Sigma(m \times 3)$ notation of [40, 43], where $|\Sigma(m \times 3)| = 3m$; the three bases thus have orders 216, 648, 1080.) The adjoints are irreducible — the 8-dimensional $\mathfrak{su}(3)$ representation does not split — by the character tables of [43]. The smaller groups $\Sigma(36 \times 3)$ (the 108-element crystallographic Valentiner analogue) and $\Sigma(60) \simeq A_5$ both have reducible adjoint at $d = 3$: the 8-dimensional $\mathfrak{su}(3)$ representation splits into smaller irreps of the subgroup, as can be read directly from the character tables [43] or verified via Eq. (6) with $\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) > 1$. These are not automatically non-universal: the Σ -invariant subspaces of $\mathfrak{su}(3)$ are not aligned with the Cartan–root decomposition (the Fourier element $V \in \Sigma(36 \times 3)$ mixes them), so even a generic diagonal extension lifts $\dim \text{Comm}(\text{Ad}_{\mathcal{S}})$ to 1 and the Sawicki–Karnas criterion is satisfied on $\mathcal{S}$. We defer $\Sigma(36 \times 3)$ and A_5 from this paper purely for compute reasons — the $t = 5/t = 10$ panel budget was set by the three irreducible-adjoint Σ -series bases — and flag a $\Sigma(36 \times 3)/A_5$ panel as natural follow-up work; the SWIFTBOT Stage-2 screen now classifies $\Sigma(36 \times 3)$ +rnd as **universal_likely** via the corrected $\dim \text{Comm}(\text{Ad}_{\mathcal{S}})$ check on generators.

The panel fixes seven extensions per base group with identical parameters across the three groups. Two

are Howard–Vala representatives of the two non-trivial Clifford-equivalence orbits at $d = 3$ (Eq. (7)): HV(1, 1, 0) (Campbell’s magic gate) and HV(2, 1, 0). Three are diagonal phase gates $P(\theta)$ at angles $\theta \in \{2\pi/9, \pi/9, 2\pi/5\}$, chosen to span both the cyclotomic-9th-root sector relevant to Section VII and the coarser $2\pi/5$ angle that generates no low-order closure against any of the three base groups. The remaining two slots per group are independent batches of ten Haar-distributed SU(3) unitaries (the *rnd* baseline). All seven extensions pass the Stage 2 universality screen (one-dimensional adjoint commutant, bounded closure open past the $|\mathcal{C}| \leq 1080$ cap).

We evaluate the panel at two design orders. The $t = 10$ pass supplies the headline Q_T values listed in Table IV and plotted in Fig. 2. The $t = 5$ pass supplies the apples-to-apples and ternary panels discussed below. $t = 10$ and $t = 5$ agree on δ to four decimals for all structured extensions except one: HV(2, 1, 0) on $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$ sits at the degenerate $\delta = 0.5000$ value at $t = 5$ (a generic small- t finite-design-order plateau, not a physical ceiling), and lifts to $\delta \approx 0.58$ at $t = 10$ once subdominant irreps of the $t=10$ ensemble become spectrally visible.

Table IV lists every $(\mathcal{C}, g)$ pair at $t = 10$, ranked within each group by δ . The rightmost column reports $Q_T / \log |\mathcal{C}| = 1 / \log(1/\delta)$, which removes the explicit $\log |\mathcal{C}|$ numerator of Eq. (3) so that rows with the same δ across different bases collapse onto a single value; this lets us read off extension-level effects without the base-group-size confound.

The panel of Table IV and Fig. 2 exhibits three clear structural features.

First, every structured extension (rational-angle $P(\theta)$ and both HV representatives, with the HV(2, 1, 0) finite-design-order plateau noted above) returns δ that is constant across $|\mathcal{C}|$ to four decimals. Because $Q_T = \log |\mathcal{C}| / \log(1/\delta)$ and $\log(1/\delta)$ is the denominator, a δ that does not depend on $|\mathcal{C}|$ forces Q_T to grow linearly with $\log |\mathcal{C}|$. Qualitatively this looks paradoxical — one might expect a bigger group to densify faster because it has more Cayley-graph vertices — but Q_T is a *word-length exponent*, not a vertex count: it measures how rapidly the Cayley random walk fills a given ε -ball as the word length grows, and when the spectral gap $1 - \delta$ is the same at every $|\mathcal{C}|$, each additional step contributes the same fractional covering, so more words are needed to compensate the extra $\log |\mathcal{C}|$ of volume to cover. The third column of Table IV, $Q_T / \log |\mathcal{C}| = 1 / \log(1/\delta)$, factors this out and collapses the structured rows onto a single value.

Second, the *rnd* extensions do not follow this pattern: the best-sample δ decreases from 0.26 on $\Sigma(72 \times 3)$ to 0.18 on $\Sigma(360 \times 3)$, so *rnd*-extended Q_T stays close to Q_{opt} rather than growing with $\log |\mathcal{C}|$. A generic Haar SU(3) draw couples to every su(3) direction, so it also activates the subdominant spectral channels that exist in the larger group but are absent in the smaller one; the structured diagonal extensions cannot reach those channels.

Third, the narrow-angle $P(\pi/9)$ row produces δ within 3% of unity and Q_T two orders of magnitude above the

rnd baseline — a $d = 3$ analog of the $14 \times$ Clifford+ T penalty at $d = 2$. Commensurability between the rational angle and the base-group orbit slows the walk drastically when the angle is finely tuned to the group structure.

The strange-state row U_S (Howard–Campbell qutrit “strange” magic state lifted to a single-qutrit unitary) is the only structured extension in our panel that behaves like a Haar draw rather than a δ -degenerate cyclotomic gate: δ tracks the *rnd* baseline within a few percent on every Σ base (0.28 on $\Sigma(72, 216 \times 3)$, 0.19 on $\Sigma(360 \times 3)$), and Q_T / Q_{opt} stays in the 1.6–2.0 band where the *rnd* entries sit. The structural reason is that U_S is a generic SU(3) rotation up to phases — it lacks the cyclotomic-root resonance that traps $P(\theta)$ rows at $\delta \sim 0.66$ –0.97. Operationally this is the most useful structured extension we test at $d = 3$: it carries a published distillation route (ternary Golay $[[11, 1, 5]]_3$ [74]) and reaches a near-*rnd* Q_T ratio without depending on a Haar draw. The strange-state rows in Table IV are at $t = 5$ only — the $t = 10$ rerun was deferred at ~ 3 h CPU on the largest base; we expect the same small lift at $t = 10$ that the HV(2, 1, 0) row exhibits.

V.1. Apples-to-apples Haar panel

To separate the base-group effect from *rnd*-extension variance, we evaluated a single batch of ten arbitrary-SU(3) extensions identically on all three groups. Table V lists the best-sample δ per matrix.

The $N = 100$ Haar panel (the first ten matrices coincide with Table V) gives mean δ values $\bar{\delta}_{\Sigma(72 \times 3)} = 0.3214$, $\bar{\delta}_{\Sigma(216 \times 3)} = 0.2623$, $\bar{\delta}_{\Sigma(360 \times 3)} = 0.2118$, with standard deviations 0.073, 0.074, and 0.090 respectively (Fig. 3). Mean δ decreases strictly monotonically with $|\mathcal{C}|$ at $\sim 5.7\sigma/\sqrt{N}$, confirming that the *rnd*-gain-with- $|\mathcal{C}|$ trend inferred from a single draw per group in Table IV survives to statistical significance. At $N = 10$ this ordering inverts between $\Sigma(216 \times 3)$ and $\Sigma(360 \times 3)$ because a single outlier ($\delta = 0.4377$ on $\Sigma(360 \times 3)$ vs. 0.24 on the smaller groups) inflates the $\Sigma(360 \times 3)$ mean by roughly one panel standard deviation — Monte-Carlo noise visible in the small panel, resolved in the large one.

The structured-panel δ -degeneracy across $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$ raises an obvious question: is this an artifact of the rational angles we happened to probe, or does it extend to arbitrary diagonal extensions of SU(3)? We scan $N = 50$ random diagonal SU(3) extensions $\text{diag}(e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3})$ (with θ_3 fixed to make $\det = 1$) on all three base groups at $t = 5$.

The degeneracy holds for every one of the 50 draws: δ on $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$ agrees bit-for-bit, while $\Sigma(360 \times 3)$ is generically distinct (per-group best/mean/max values across the 50 draws are 0.4516/0.6769/0.9921 on the paired $\Sigma(72 \times 3)$ – $\Sigma(216 \times 3)$ and 0.2539/0.5106/0.9662 on $\Sigma(360 \times 3)$). The pattern is thus not cyclotomic — it is the $\Sigma(72 \times 3) \subset \Sigma(216 \times 3)$ nested-subgroup structure [43]: at $t = 5$, the dominant t -design irreps of SU(3) restrict

TABLE IV. $d = 3$ survey at $t = 10$, ranked within each group by δ . Rnd rows report the best δ across two independent Haar draws. Strange-state (U_S) rows are at $t = 5$ only and marked †; the $t = 10$ rerun was deferred (one-shot, but ~ 3 h CPU on the largest base). Kesten–McKay floor Q_{opt} : 2.692, 2.544, 2.495 for $\Sigma(72, 216, 360 \times 3)$.

Group	Extension	δ	Q_T	$Q_T / \log \mathcal{C} $
$\Sigma(72 \times 3)$	rnd	0.2605	3.996	0.744
	$U_S^\dagger$	0.2845	4.276	0.797
	HV(2, 1, 0)	0.5927	10.277	1.911
	$P(2\pi/5)$	0.6545	12.681	2.360
	HV(1, 1, 0)	0.6764	13.750	2.559
	$P(2\pi/9)$	0.8830	43.208	8.041
	$P(\pi/9)$	0.9698	175.561	32.671
$\Sigma(216 \times 3)$	rnd	0.2107	4.157	0.643
	HV(2, 1, 0)	0.5833	12.011	1.856
	$U_S^\dagger$	0.2845	5.150	0.797
	$P(2\pi/5)$	0.6545	15.273	2.360
	HV(1, 1, 0)	0.6764	16.561	2.559
	$P(2\pi/9)$	0.8830	52.039	8.041
	$P(\pi/9)$	0.9698	211.443	32.671
$\Sigma(360 \times 3)$	$U_S^\dagger$	0.1948	4.270	0.611
	rnd	0.1784	4.052	0.579
	HV(2, 1, 0)	0.5698	12.419	1.775
	$P(2\pi/5)$	0.6545	16.478	2.360
	HV(1, 1, 0)	0.6764	17.867	2.559
	$P(2\pi/9)$	0.8830	56.145	8.041
	$P(\pi/9)$	0.9698	228.127	32.671

TABLE V. Apples-to-apples Haar panel at $t = 5$: ten pre-generated single-qutrit extensions evaluated identically against each of the three base groups. The smallest δ in each row is bolded. Mean δ over the ten matrices appears in the last row; lower is better.

matrix	δ		
	$\Sigma(72 \times 3)$	$\Sigma(216 \times 3)$	$\Sigma(360 \times 3)$
apple00	0.2974	0.2974	0.3462
apple01	0.4636	0.4636	0.2029
apple02	0.2221	0.1727	0.2285
apple03	0.2493	0.2292	0.2299
apple04	0.3430	0.2104	0.1197
apple05	0.2446	0.2412	0.4377
apple06	0.2842	0.1701	0.1965
apple07	0.3169	0.2312	0.2271
apple08	0.3996	0.2377	0.3795
apple09	0.2526	0.2526	0.1823
mean	0.3073	0.2506	0.2550

to representations that both groups act on identically, so every diagonal extension couples through the same spectral channel on both. $\Sigma(360 \times 3)$ is not a supergroup of either (it contains A_5 rather than the $\Sigma(72 \times 3)$ center-extended chain), hence its distinct δ . A $t = 10$ rerun would test whether a subdominant irrep of $\Sigma(216 \times 3)$ absent in $\Sigma(72 \times 3)$ becomes spectrally visible and lifts the degeneracy.

Fig. 4 plots the full distribution on the $SU(3)$

maximal-torus simplex; each random-diagonal point is S_3 -symmetrized to six positions with the same δ because the qutrit Clifford subgroup contains the Weyl permutations.

The best $(\mathcal{C}, g)$ pair found in the survey is the **apple04** rnd matrix on $\Sigma(360 \times 3)$ at $t = 5$, with $\delta = 0.1197$ and $Q_T = 3.291$ — within 32% of $Q_{\text{opt}} = 2.495$. The best deterministic extension is HV(2, 1, 0) on $\Sigma(72 \times 3)$, giving $Q_T = 10.28$ at $t = 10$, a factor 3.8 above $Q_{\text{opt}} = 2.692$. The ranking order at $t = 5$ and $t = 10$ matches for every row except the HV(2, 1, 0) floor lift already discussed; the four-decimal cross-group δ -degeneracy and its ordering consequences are therefore not a finite- t artifact but a structural feature of the rational-angle and Howard–Vala extensions on this Σ -series.

The rnd winners like **apple04** are generic Haar draws, not algebraic unitaries over a known ring, and off-the-shelf magic-state distillation protocols target structured states: $|T\rangle$ via Bravyi–Kitaev / Bravyi–Haah triorthogonal [2, 3], rational-angle rotations $|\psi(\theta)\rangle$ via programmable MSD on a Clifford+ T substrate [58–60], or cyclotomic phases via code-switching to a permutation-invariant (PI) code [5]. We are not aware of any published protocol that certifies fault-tolerant preparation of a generic Haar $SU(d)$ magic state; the Q_T figure therefore ranks approximation reachability, not fault-tolerant cost, and the rnd winners of Table IV should be read as upper bounds on what a structured near-optimal extension could achieve — open on both the algebraic-synthesis and code-distillation sides.

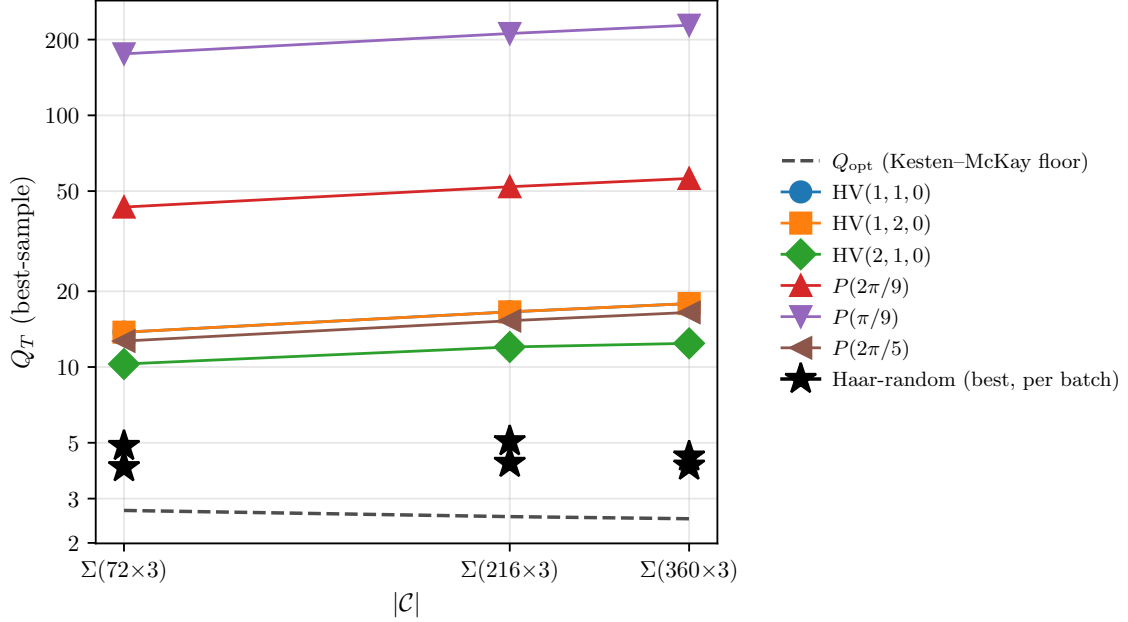


FIG. 2. Q_T across the three Σ -series base groups at $d = 3$, $t = 10$, for every extension in Table IV. The dashed line is the Kesten–McKay floor Q_{opt} . Structured extensions (rational-angle $P(\theta)$ and Howard–Vala) run nearly parallel to it at a factor ~ 3 –70 offset; best-random extensions (stars) approach Q_{opt} as $|\mathcal{C}|$ grows.

VI. $d = 4$ FAIRBAIRN PILOT

To probe whether the $Q_T/Q_{\text{opt}} \rightarrow 1$ trend observed at $d = 2, 3$ continues at $d = 4$, we ran a small Fairbairn-subgroup pilot at low design order. Compute is the binding constraint: the largest irreps scale as t^6 at $d = 4$ and matrix storage as t^{12} , so the pilot stops at $t = 4$ on the four smallest groups (A_5 , $\Sigma(60 \times 4)$, $\Sigma(120 \times 4)_{1,2}$) and at $t = 2$ on $\Sigma(720 \times 4)$ and S_7 . The extension panel is a four-angle diagonal grid $P(\theta)$ with $\theta \in \{0.0625, 0.125, 0.25, 0.4\}$ rad (chosen as a generic test grid, not paper-aligned cyclotomic angles), a two-mode *angles* extension at the same scale, and a single arbitrary-SU(4) (*rnd*) draw at $t \in \{3, 4\}$ on the four small groups. Table VI reports the best-sample Q_T and ratio to the Kesten–McKay floor for each pair.

Two observations carry over from the $d = 2, 3$ surveys. (i) Structured small-angle $P(\theta)$ extensions sit at $Q_T/Q_{\text{opt}} \in [10, 1000]$ depending on how finely the angle resonates with the base-group orbit — the same commensurability slowdown seen for Clifford+ T at $d = 2$ and $\Sigma + P(\pi/9)$ at $d = 3$, now amplified by the extra small angles in this test grid. (ii) The single arbitrary-SU(4) draws settle in the $Q_T/Q_{\text{opt}} \in [1.0, 1.9]$ range, qualitatively matching the Haar super-golden behaviour at $d = 2, 3$. Most striking: $A_5 + \text{rnd}$ at $t = 3$ gives $Q_T = 3.007$, ratio 1.0006 to the floor $Q_{\text{opt}} = 3.005$ — essentially saturating Kesten–McKay on a single sample; the $t = 4$ rerun lifts this to ratio 1.12, indicating the saturation is a low-design-order artefact rather than a converged number, but the closeness already suggests A_5 is a strong $d = 4$ super-golden candidate. The $\Sigma(120 \times 4)_2$

TABLE VI. $d = 4$ Fairbairn-subgroup pilot: best-sample Q_T per $(\mathcal{C}, g)$ pair at the largest design order achieved. “s.” columns indicate the t -design depth reached. Columns under $P(\theta)$ are diagonal phase gates at the four angles θ (rad); “2-ang” is the 2-angle diagonal at $\theta_1 = 0.25$; “rnd” is a single arbitrary-SU(4) draw at $t = 3$ then $t = 4$. Q_T/Q_{opt} is shown in parentheses for the rnd column. “—” marks pairs not run.

$\mathcal{C}$	$ \mathcal{C} $	Q_{opt}	0.0625	0.125	0.25	0.4	rnd ($t=3, t=4$)	
A_5	60	3.01	1703	428	110	45	3.01 (1.00), 3.37 (1.12)	
$\Sigma(60 \times 4)$	240	2.67	948	237	60	24	4.58 (1.71), 4.92 (1.84)	
$\Sigma(120 \times 4)_1$	480	2.58	1068	267	67	27	3.92 (1.52), 4.42 (1.71)	
$\Sigma(120 \times 4)_2$	480	2.58	2567	646	166	68	4.68 (1.82), 4.96 (1.92)	
$\Sigma(720 \times 4)$	2880	2.42	1032	258	64	25	—	
S_7	5040	2.39	1105	276	68	26	—	

pair at $t = 3, 4$ also lands in the 1.8–1.9 ratio band, comparable to $\Sigma(360 \times 3)$ ’s 1.32 at $d = 3$. All Fairbairn rows in Table VIII carry $\times$ in every fault-tolerance column.

VII. THE DSA PRIMITIVE-GATE TARGET FAMILY

Many applications do not need to approximate every unitary in $\text{SU}(d)$; they need a *finite, enumerable* list of target unitaries — a structured workload for which the cost of interest is the mean synthesis depth over that target list, not a Haar-uniform covering rate. On such a workload the right question is not “which $(\mathcal{C}, g)$ maximises the spectral gap?” but “which $(\mathcal{C}, g)$ synthesises this specific target

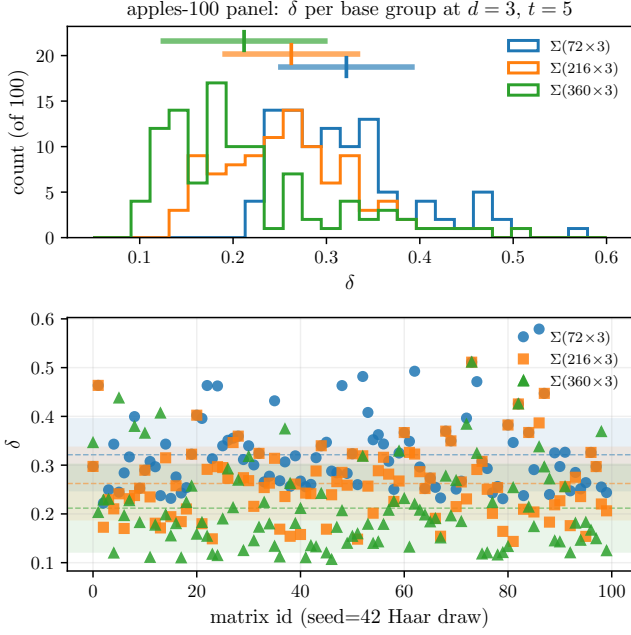


FIG. 3. Apples-to-apples Haar panel at $N = 100$ matrices on the three $d = 3$ Σ -series base groups. Top: overlaid δ histograms. Bottom: per-matrix δ for each group; the first ten matrices coincide with Table V. The mean δ decreases strictly monotonically with $|\mathcal{C}|$ at $\sim 5.7\sigma/\sqrt{N}$; the single outlier visible as the $\Sigma(360 \times 3)$ peak near $\delta \simeq 0.44$ is the one that inverted the ten-matrix ordering in Table V.

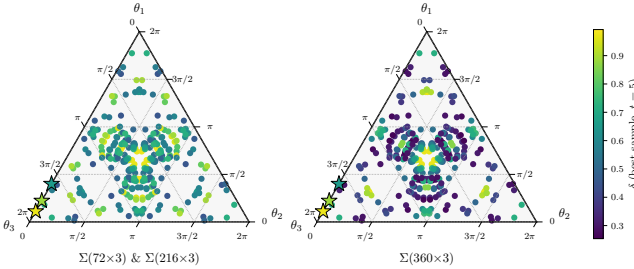


FIG. 4. δ on the $SU(3)$ maximal torus at $t = 5$. Each panel shows the 50 random-diagonal extensions, S_3 -symmetrized to six copies each (300 points per panel). Left: δ values that are bit-for-bit identical on $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$; right: $\Sigma(360 \times 3)$. Red stars mark the three structured rational-angle extensions $P(2\pi/9)$, $P(\pi/9)$, $P(2\pi/5)$, colored by their own δ ; they sit on the $\theta_1 = 0$ edge because each has the form $\text{diag}(1, e^{i\theta}, e^{-i\theta})$.

list with the shortest circuits?” The answers disagree in general, and the overhead difference can be large enough to motivate changing the extension or even the base group for a given workload, as we show below. This in turn feeds back to the fault-tolerance stack: an extension or group that delivers a structured workload cheaply is worth supporting with new QEC and magic-state protocols even if it ranks poorly on Q_T . Discrete-subgroup approximations of lattice gauge simulations [29, 48, 61] provide a concrete such workload: a few continuous-angle R_Z rotations with

group-theoretic coefficients, plus a finite set of cyclotomic phases from the discrete Fourier transform U_F . We note that $\Sigma(36 \times 3)$ itself appears here as a *target family* of $SU(2)$ rotations derived from its trace spectrum under a qubit encoding [30]; it is not used as a base group, its reducible adjoint at $d = 3$ having excluded it from the QCO survey (Section V).

A concrete target family. We work out the case of the $\Sigma(36 \times 3)$ encoding as a representative example; different subgroups in the series produce different coefficient sets. The trace gate $U_{\text{Tr}}(\theta) = \text{diag}_{g \in \mathcal{C}}(e^{i\theta \text{Re Tr}(g)})$ appears in the magnetic plaquette of any DSA lattice gauge simulation [29]. For $\Sigma(36 \times 3)$, Gustafson *et al.* [30] reduce it to a three-qubit trace register: the 14 conjugacy classes collapse to 7 distinct $\text{Re Tr}(g) \in \{3, \pm 1, \pm \frac{1}{2}, -\frac{3}{2}, 0\}$, and a Boolean map $(p, q, r, s, t) \mapsto (v_0, v_1, v_2)$ separates them. On the (v_0, v_1, v_2) register the trace Hamiltonian factorizes as

$$H_{\text{Tr}} = \frac{1}{16} [3(III + IIZ + IZI) + 5 IZZ + ZII + 9(ZIZ + ZZI) + 15 ZZZ], \quad (8)$$

so $U_{\text{Tr}}(\theta)$ decomposes (up to Clifford conjugation) into R_Z rotations at angles proportional to $\alpha_c \theta$ for $\alpha_c \in \{1, 3, 5, 9, 15\}$ — the five distinct non-zero coefficients in the decomposition. These coefficients are specific to $\Sigma(36 \times 3)$; other subgroups in the series produce different trace spectra and different α_c . The discrete half comes from U_F : up to global phase, the Howard–Vala qutrit gate $HV(1, 2, 0)$ [38] injects 9th-root phases, requiring $R_Z(2\pi k/9)$ for $k = 1, \dots, 8$. The synthesis target for the $\Sigma(36 \times 3)$ qubit pipeline is therefore

$$\mathcal{T}_{36} = \{R_Z(2\pi k/9)\}_{k=1}^8 \cup \{R_Z(\alpha_c \theta)\}_{\alpha_c \in \{1, 3, 5, 9, 15\}, \theta \in (0, \pi)}. \quad (9)$$

Mean-depth figure of merit. Given a qubit base group $\mathcal{C}$ and single-gate extension g , SWIFTBOT enumerates the word tree of $\mathcal{C} \cup \{g, g^{-1}\}$ by breadth-first search with projective-phase deduplication, capped at 2×10^5 unique words. For each $U \in \mathcal{T}_{36}$ we record the shortest BFS word whose projective distance $d(U, W) = \sqrt{2 - |\text{Tr}(U^\dagger W)|}$ falls below $\varepsilon = 10^{-2}$, or the depth of closest approach if no such word exists. The $\varepsilon = 10^{-2}$ floor matches the per-link accuracy typical of near-term Trotter budgets [29]; the BFS-exhaust cap prevents runaway search at this tolerance, and tightening ε can only increase the $6.8\times$ advantage (see §Caveats below). Discrete targets may be hit *exactly* at finite depth, in which case the corresponding distance is zero and the T -count is read directly from the word. Continuous targets are sampled at ten uniform $\theta \in (0, \pi)$ per α_c , for fifty continuous evaluations per extension. We report mean distance across Eq. (9) as the single headline number; ties are broken on the discrete-exact count.

Result. Table VII ranks the candidates at BFS depth 15, $\varepsilon = 10^{-2}$, $N_\theta = 10$ per continuous coefficient. The Clifford+ $P(2\pi/9)$ extension attains mean projective distance 0.0122 on $\mathcal{T}_{36}$ with 27 of 58 targets hit below ε ,

TABLE VII. DSA $\Sigma(36 \times 3)$ target-family coverage at BFS depth 15, $\varepsilon = 10^{-2}$, 10 continuous θ -samples per α_c . “hits” counts the 58 targets (the 8 discrete 9th-root phases in Eq. (9) plus 50 continuous $R_Z(\alpha_c\theta)$ samples) reached below ε . “mean/+max” are projective distances across all 58; lower is better. $\langle T \rangle$ is the mean T -count along hit words. “method” is **rs** for Ross–Selinger [10] exact synthesis and **bfs** for the SWIFTBOT BFS estimate; all rows are **bfs** except the canonical $P(\pi/4)$, no RS-style normal form having yet been established for the rational-angle cyclotomic rings $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ or $\mathbb{Z}[i, \sqrt{2}]$.

Base + ext.	hits (/58)	mean dist.	max dist.	$\langle T \rangle$	method
Clifford+ $P(\pi/18)$	34	0.0104	0.0420	2.79	bfs
hurwitz + $P(\pi/18)$	35	0.0112	0.0420	5.37	bfs
Clifford+ $P(\pi/9)$	27	0.0120	0.0401	2.04	bfs
Clifford+ $P(2\pi/9)$	27	0.0122	0.0401	2.11	bfs
Clifford+ $P(\pi/8)$	7	0.0221	0.0387	0.29	bfs
Clifford+ T_{24}	6	0.0228	0.0465	0.83	bfs
Clifford+ T_{12}	8	0.0233	0.0474	1.88	bfs
hurwitz + T_{12}	0	0.0313	0.0573	—	bfs
Clifford+ $P(\pi/4)$	5	0.0833	0.1554	0.00	rs

while the canonical Clifford+ T extension attains mean distance 0.0833 with only 5 hits—a factor $6.8\times$ separation in mean distance. The equivalence class of $P(2\pi k/18)$ with $\gcd(k, 18) = 1$ (which includes $P(\pi/9)$ and $P(\pi/18)$) all saturate the discrete 9th-root targets, a direct consequence of $\langle \text{Clifford}, P(2\pi/9) \rangle$ closing at a finite group of $\sim 3 \times 10^4$ elements; the algebraic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ supplies every 9th-root phase via a short word. Within this equivalence class, Clifford+ $P(\pi/18)$ marginally edges Clifford+ $P(2\pi/9)$ at depth 15 (mean 0.0104 vs. 0.0122; 34 vs. 27 hits), reflecting the strictly larger 18th-root coverage: $P(\pi/18)$ generates ω_{36} , which contains both the 9th- and 18th-root phases of U_F . Crucially, the super-golden extensions T_{24} and T_{12} —which optimize Q_T for Clifford and for the 12-element projective A_4 base **hurwitz** respectively [47]—perform *worse* on $\mathcal{T}_{36}$ than the rational-angle rows: their algebra $\mathbb{Z}[i, \sqrt{2}]$ does not contain ω_9 , so they cannot reach the 9th-root targets exactly at any depth; on the $|\mathcal{C}| = 12$ **hurwitz** base the T_{12} extension hits zero targets at depth 15. The **hurwitz**+ $P(\pi/18)$ row is competitive (35 hits, mean 0.0112) and does so with a strictly *smaller* base group: in the target-family regime the overhead advantage tracks the cyclotomic content of the extension, not the size of the base.

Interpretation. Clifford+ $P(2\pi/9)$ is worse on Q_T than Clifford+ T yet $6.8\times$ better on mean depth over $\mathcal{T}_{36}$. The reason is structural: Q_T is the coefficient of a Haar-uniform covering rate, while $\mathcal{T}_{36}$ is a measure-zero algebraic slice of $\text{SU}(2)$ whose ω_9 cyclotomic content matches $\langle \text{Clifford}, P(2\pi/9) \rangle$. The super-golden extensions saturate Q_{opt} but equidistribute in the wrong direction for this target. A lattice-gauge simulator compiling Eq. (9) should therefore rank extensions by target-family mean depth, not by Q_T , and the two rankings can disagree by nearly an order of magnitude.

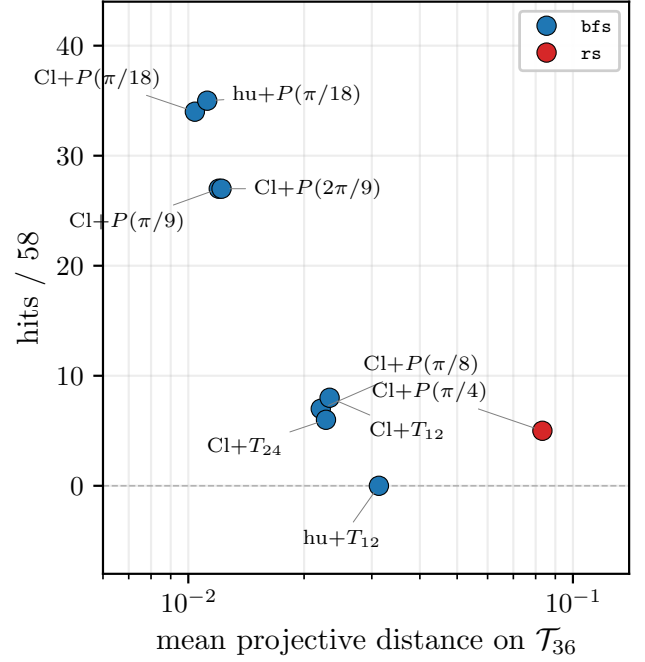


FIG. 5. DSA $\Sigma(36 \times 3)$ target-family coverage from Table VII: hits/58 versus mean projective distance on $\mathcal{T}_{36}$ at BFS depth 15, $\varepsilon = 10^{-2}$. Canonical Clifford+ T (red, **rs_exact** certified) sits bottom-right with the largest mean distance and only 5 hits. Rational-angle $P(k\pi/18)$ extensions (blue, **bfs_estimate**) cluster top-left with many hits at an order of magnitude smaller distance; super-golden T_{12}/T_{24} rows fall in between despite saturating Q_{opt} (Section IV). The visual separation matches the $6.8\times$ headline.

Caveats. The Clifford+ T row’s T -count is certified by Ross–Selinger exact synthesis [10, 11]; the companion Clifford-cyclotomic framework of Forest *et al.* [62] extends this to $\mathbb{Z}[\zeta_{2^n}]$ but not to the rational-prime-denominator rings relevant here. No analogous normal form exists for $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$, so the rational-angle rows carry depth-15 BFS upper bounds. The $6.8\times$ advantage is therefore a *lower bound*: a certified synthesis theorem over $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ would only shorten the BFS-side words, while the canonical Ross–Selinger side is already tight. Anderson–Jochym–O’Connor [18] excludes any qubit *stabilizer* code with transversal $|k\pi/9\rangle$. Two routes sidestep this. Programmable MSD [58–60] is indirect synthesis: it prepares $|\psi(\theta)\rangle$ from already-distilled $|T\rangle$ on a Clifford+ T stabilizer substrate, so the AJOC no-go never applied to it. Non-stabilizer code-switching to a Kubischta–Teixeira PI code [5, 63] is a genuine bypass but requires a PI code of sufficient distance hosting $|\omega_9\rangle$, which has not been constructed. The $6.8\times$ advantage is cleanest when $P(2\pi/9)$ is a native primitive—e.g., in a qutrit register or on a hypothetical nonadditive code that distills $|\omega_9\rangle$ directly.

VIII. DISCUSSION

Two findings drive the discussion below. First, at $d = 3$ and $t = 5$, arbitrary-SU(3) extensions on the largest irreducible-adjoint Σ -series base reach $Q_T/Q_{\text{opt}} \approx 1.32$ —essentially the same “golden-gate direction” observed at $d = 2$, where the arbitrary-SU(2) baseline sits 4% above Q_{opt} on the 12-element hurwitz base (Section IV). No structured extension in our panel (Howard–Vala or rational diagonal) gets closer than a factor ~ 2.9 of Q_{opt} . Second, and in apparent tension with the first, the non-canonical Clifford+ $P(2\pi/9)$ extension beats the canonical Clifford+ T by $6.8\times$ on the DSA primitive-gate target family (Section VII) despite being $\sim 8\times$ worse on Q_T . The two findings together are the paper’s central design lesson: the “best general-purpose extension” and the “best target-specific extension” optimize different objectives, and systematic qudit fault-tolerance design must report both.

VIII.1. Fault-tolerance status of the winning extensions

A Q_T ranking is necessary but not sufficient for fault-tolerant implementation. Table VIII audits every extension evaluated against the curated SWIFTBOT catalogs of transversal codes, distillation protocols, and certified synthesis rings. Only canonical Clifford+ T at $d = 2$ admits an end-to-end certified stack and serves as our validation anchor. Three rows inherit a transversal-code half only: hurwitz-based super-golden extensions via Parzanchevski–Sarnak [54], BI+rnd via the Kubischta 2I code [6], and the $d = 3$ $\Sigma + \text{HV}(z, \gamma, \varepsilon)$ rows via the qutrit triorthogonal $[[9m - k, k, 2]]_3$ family [23]. All other rows lack either a distillation circuit or a certified synthesis ring.

Three regimes deserve comment. First, the cyclotomic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ —hosting the $d = 2$ target-family winners $P(2\pi/9)$, $P(\pi/9)$, $P(\pi/18)$ —has no Ross–Selinger-style normal form, and the two non-stabilizer handling routes discussed in Section VII Caveats (indirect programmable MSD on a Clifford+ T substrate; PI code-switching pending a suitable distance-preserving PI code) remain the best-known options at $d = 2$. At $d = 3$ the same rotations are qutrit-native and Campbell–Anwar–Browne [23] distills $|\omega_9\rangle$ directly.

Second, the super-golden gates of Parzanchevski–Sarnak [54] live in $\mathbb{Z}[i, \sqrt{2}]$ and are not Clifford-orbit rotations of stabilizer states, so triorthogonal distillation does not apply. Whether a golden-gate-compatible distillation circuit can be built—possibly on the Kubischta 2I code, whose transversal group is precisely the super-golden base—is open.

Third, arbitrary-SU(d) extensions are by definition not exactly synthesized, and so cannot be distilled in the usual sense [2, 3]. This applies uniformly to every rnd row; the best current claim is transversal-group support

at the base-code level.

For the Σ -series at $d = 3$, data and distillation codes are structurally separable, and the transversal side is partially solved: Kubischta and Teixeira [63] explicitly construct a family of qutrit codes with transversal $\Sigma(360 \times 3)$ action via twisted unitary t -groups, the smallest being a $[[7, 1, 2]]_3$ nonadditive code (the χ_3 irrep family at $n \equiv 1 \pmod 3$, $n \geq 7$) and an alternative $[[5, 1, 2]]_3$ from the χ_4 family ($n \equiv 2 \pmod 3$, $n \geq 5$). Both detect any single-qutrit error and realize the full 1080-element Valentiner group transversally, so our $\Sigma(360 \times 3) + \text{rnd}$ winner has a data-code partner (Table VIII). Analogous constructions for $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$ remain open; so does a distance-three member of the $\Sigma(360 \times 3)$ family. On the distillation side, the Howard–Vala magic state itself is distilled by the qutrit triorthogonal family [23, 24]. Denys–Leverrier covariant codes at $d = 2$ [7, 8] support $2T$ and $2O$ group actions natively through continuous-group covariance, distinct from Eastin–Knill transversality on a stabilizer code. (Clifford+ R synthesis [45] separately closes the compilation-certification gap for these $d = 3$ gate sets; we return to this below.)

Higher-dimensional qudit MSD is advancing rapidly. Krishna–Tillich [64] opened the low-overhead program; Wills, Hsieh, and Yamasaki [25] achieved the first constant-overhead MSD construction via asymptotically good quantum codes; Golowich–Guruswami [65] and Nguyen [66] gave asymptotically good qubit codes with transversal non-Clifford and CCZ gates. Most directly applicable to our framework, Cervia *et al.* [26] construct triorthogonal codes on $\mathbb{F}_{2^{2m}}$ at qudit dimensions as small as $q = 64$ that beat the Tsfasman–Vlăduț–Zink bound and give constant-space-overhead $|CCZ\rangle$ distillation, together with an explicit qudit-to-qubit conversion circuit that makes the construction a drop-in upgrade for any Clifford+CCZ evaluation. None of these high- d codes have cataloged transversal pairings with our base-group registry, but the infrastructure now exists to pair them.

A complementary recent development is *magic-state cultivation* [4]: rather than distilling high-fidelity $|T\rangle$ states from many noisy copies, one grows a protected $|T\rangle$ inside a surface-code patch using roughly the same gate budget as a logical CNOT, reducing qubit-round cost by an order of magnitude at realistic noise [27], with an experimental demonstration on a superconducting processor [28]. Cultivation is currently qubit-native and surface-code-specific; the open question for the qudit program is whether a native- d analogue exists on codes with transversal non-Clifford structure (Kubischta 2I, Denys–Leverrier $2T/2O$, or the twisted t -group codes of [63]). Two arguments have historically favored $d = 2$: the maturity of Ross–Selinger Clifford+ T synthesis and the qubit-specificity of magic-state cultivation. The first is closing at $d = 3$ with Clifford+ R synthesis [45]; a qudit cultivation analogue would close the second.

On the synthesis side of the qudit story, single-qutrit exact-synthesis methods over Clifford+ T and related Clifford+cyclotomic rings are by now well developed [45, 67–

[69], so the qutrit compilation story is closing to parity with the qubit one.

Our $t = 10$ result that $\Sigma(216 \times 3) + \text{HV}(1, 1, 0)$ gives $Q_T = 16.56$ on the qutrit Clifford base — a factor ~ 6.5 above the Kesten–McKay floor $Q_{\text{opt}} = 2.54$ — is a numerical shadow of a sharper number-theoretic result: Evra and Parzanchevski [70] prove that the qutrit Clifford+ T gate set generates a *thin* matrix group in $\text{PU}(3)$ (Zariski-dense but of infinite index in its ambient S -arithmetic group), so no near-optimal covering theorem applies and the covering rate is bounded away from the golden-gate exponent. Their Clifford+ D extension — replacing the single T -gate with the diagonal subgroup $\mathcal{D} \cong (\mathbb{Z}/18)^3 \subset \text{U}(3)$ of signed 9th-root diagonals ($|\mathcal{D}| = 5832$ in $\text{U}(3)$, or 324 distinct elements in $\text{PU}(3)$ after quotienting by the common phase) — recovers arithmeticity and covers $\text{PU}(3)$ within a factor $\log_3 105 \approx 4.24$ of optimal; and arithmetic-lattice constructions on Bruhat–Tits buildings [71] saturate the bound but live in infinite groups. Both are outside the single-gate $(\mathcal{C}, g)$ framework of this paper: Clifford+ $\mathcal{D}$ requires a set-valued generalization $(\mathcal{C}, \mathcal{G})$, and the arithmetic-lattice golden gates are not finite subgroups at all.

VIII.2. Limitations

Three limitations frame the numerical claims. First, the $d = 3$ Haar statistics come from either single-sample $t = 10$ draws or the 100-matrix apples-to-apples panel at $t = 5$ (Section V.1); a joint $t = 10$, ~ 100 -sample Haar run is held up by a qco-kernel limitation that aggregates each subprocess to a single output row regardless of sample size. Second, the $d = 4$ Fairbairn pilot (Section VI) reaches only $t = 4$ on the four smallest groups and $t = 2$ on $\Sigma(720 \times 4)$ and S_7 ; largest-irrep dimensions scale as t^6 at $d = 4$ and matrix storage as t^{12} , so a converged $t = 10$ run on the large- $|\mathcal{C}|$ $\text{SU}(4)$ subgroups would need a cluster slot. Third, the target-family T -counts in Table VII are depth-15 BFS upper bounds except for the Ross–Selinger-certified Clifford+ $P(\pi/4)$ row. The $6.8\times$ advantage is a *lower bound*: a certified normal form over $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ [62] would only tighten the BFS side, not the canonical side, which is already exact.

VIII.3. Open questions

Two quantitative questions are directly posed by our $t = 5$ data and should be sharp enough to settle at $t \geq 10$: (a) does the four-decimal δ -degeneracy across $\Sigma(216, 648, 1080 \times 3)$ observed for structured diagonal and Howard–Vala extensions in Table IV persist, break, or re-rank at larger t ? and (b) does the $\Sigma(360 \times 3)$ apples-panel mean cross strictly below the $\Sigma(216 \times 3)$ mean once the ~ 100 -matrix apples panel removes the 10-matrix outlier sensitivity? Question (b) has been *settled in the affirmative* at $t = 5$ in Section V.1 ($5.7\sigma/\sqrt{N}$ separation at

$N = 100$); the $t \geq 10$ rerun will establish whether the same ordering holds at converged design order. Three qualitative questions are left open: (c) a super-golden construction at $d = 3$ saturating $Q_{\text{opt}} \approx 2.69$ on $\Sigma(72 \times 3)$, 2.54 on $\Sigma(216 \times 3)$, or 2.50 on $\Sigma(360 \times 3)$; (d) an explicit data code C_{data} that hosts one of our Σ -series base groups transversally, analogous to the Denys–Leverrier constructions for $2T$ and $2O$ at $d = 2$ [7, 8] or to the Kubischta–Teixeira twisted unitary t -group framework [63]; and (e) a Ross–Selinger-style normal form for the cyclotomic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ that would certify the Table VII T -counts.

VIII.4. Software release

SWIFTBOT is released with this paper [49] under the MIT license. The release contains: the five-stage pipeline (Section III); the pre-seeded group registry for $d \in \{2, 3, 4\}$; the SWIFTBOT optimizations of the QCO kernel of [46], including the conjugation cache and on-disk $\pi_\lambda(\mathcal{C})$ cache described in Section III; the Stage-4 QEC-code catalog and Stage-5 distillation-protocol catalog with their cross-references to transversal groups; the BFS target-family coverage module used in Section VII; the `q-panel` CLI entry point and the two panel specifications `d3_survey` and `d3_rnd_apples` that reproduce Tables IV and V bit-identically; the SQLite cache containing all 54 per-sample rows quoted in Section V; and the L^AT_EX source of this paper. A minimal README walks new users through reproducing Tables II, IV and V on a laptop in under an hour. Pull requests adding new base groups, extensions, codes, or distillation protocols are welcome; the catalog cross-validation tests ensure that every registered transversal-group string resolves to a registered base group.

IX. CONCLUSIONS

SWIFTBOT—an agentic workflow that ranks $(\mathcal{C}, g)$ pairs by Q_T and cross-references each against a curated fault-tolerance catalog—produces the first systematic $d = 3$ Q_T ranking in our registry and identifies $\Sigma(360 \times 3)$ with an arbitrary- $\text{SU}(3)$ extension as the best general-purpose qudit base ($Q_T/Q_{\text{opt}} \approx 1.32$ in the apples-100 panel at $t = 5$, and consistent single-sample draws at $t = 10$). A systematic 10^4 -sample arbitrary- $\text{SU}(2)$ pilot on the binary icosahedral base at $d = 2$ gives $Q_T/Q_{\text{opt}} \approx 1.27$, the natural super-golden candidate. Applied to a lattice-gauge simulation target family, the non-canonical Clifford+ $P(2\pi/9)$ beats canonical Clifford+ T by $6.8\times$ on mean depth despite being worse on Q_T , establishing that the two rankings locate different gates. Only canonical Clifford+ T admits an end-to-end certified fault-tolerant stack; every other winner is covered partially, with sharp open algebraic-compilation or code-pairing problems standing between it and full fault-tolerance. Four concrete next steps carry most of the remaining

design risk. (i) A $d = 4$ pilot on the Fairbairn subgroups [40] would test whether the $Q_T/Q_{\text{opt}} \rightarrow 1$ trend continues past $d = 3$ or the $d = 3$ “golden direction” is an accident. (ii) An explicit Σ -series $\leftrightarrow$ data-code transversal pairing [5, 8, 63] would upgrade the $d = 3$ winners from PARTIAL to COMPLETE in Table VIII. (iii) A Ross–Selinger normal form for $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ would replace the BFS upper bound in Table VII and likely sharpen the $6.8\times$ headline. (iv) A *set-valued* generalization of the framework from single-gate extensions $(\mathcal{C}, g)$ to gate-set extensions $(\mathcal{C}, \mathcal{G})$ would bring constructions like the Evra–Parzanchevski Clifford+ $\mathcal{D}$ PU(3) golden gates [70] into the Q_T ranking — empirically testing whether the $\log_3 105$ covering-rate factor is recovered on $\Sigma(216\times 3)+\mathcal{D}$, and plausibly exposing other set-valued near-optima that single-extension Q_T cannot see. For a hardware vendor choosing between qubit and qudit architectures *today*, the

practical takeaway is narrower: if the workload is lattice-gauge-simulation-shaped, a rational-phase qubit Clifford extension already buys nearly an order of magnitude over the canonical Clifford+ T stack with no new hardware.

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